

Stock Price Prediction using PySpark

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Abstract

This report details an investigation into predicting stock market trends using advanced time series forecasting techniques, specifically focusing on historical price data for Apple Inc. (AAPL). The project employs a modern data pipeline approach, leveraging PySpark for scalable data ingestion, cleaning, and the engineering of relevant features such as moving averages. For the core predictive modeling, a Long Short-Term Memory (LSTM) neural network, recognized for its capability in handling sequential data and capturing temporal dependencies, is utilized. The methodology encompasses transforming the time series data into the requisite sequence format for LSTM input, training the model using Keras/TensorFlow, and rigorously evaluating its predictive performance. Key evaluation metrics including Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) are presented, complemented by a visual analysis comparing the model's price predictions against the actual historical movements. This work demonstrates the effective synergy between big data processing tools and deep learning architectures for tackling complex financial forecasting challenges.

Contents

1	Introduction	3
2	Data Acquisition and Preparation	3
2.1	PySpark Data Pipeline	3
2.2	Transition to Deep Learning Environment	4
2.3	Data Scaling and Sequence Generation for LSTM	4
3	LSTM Model: Architecture and Training	4
3.1	Model Architecture	4
3.2	Model Compilation and Training	5
4	Results and Performance Evaluation	5
5	Conclusion and Future Directions	6

1 Introduction

Predicting stock market movements represents a classic and complex challenge within financial time series analysis. The inherent volatility, noise, and influence of numerous unpredictable external factors make this a difficult forecasting problem. While traditional statistical methods provide valuable insights, the highly non-linear and long-term dependencies present in financial data often necessitate more powerful modeling techniques. Deep learning, particularly recurrent neural networks like Long Short-Term Memory (LSTM), has emerged as a promising approach due to their ability to learn from sequences and maintain memory over time.

Furthermore, handling the vast amounts of historical and real-time financial data streams underscore the need for efficient and scalable data processing solutions. Big data frameworks, such as Apache Spark with its Python API (PySpark), offer the capabilities required for ingesting, cleaning, transforming, and engineering features from large-scale historical data, serving as an essential preparatory step for sophisticated modeling.

This project investigates the application of an LSTM network for forecasting daily stock closing prices, specifically for Apple Inc. (AAPL), by integrating a PySpark-based pipeline for initial data handling and feature engineering. The aim is to build a robust forecasting model and analyze its performance on real-world financial time series data, showcasing competence in both big data analytics and deep learning techniques within the context of financial markets.

This research was conducted as part of the CAS7A1 course on Big Data Analytics (Spring 2016) in the M.Sc. Computer Science program at the National Institute of Technology, Tiruchirappalli, under the esteemed guidance of Prof. Nicholas.

2 Data Acquisition and Preparation

The dataset used for this project consists of historical daily price and volume data for Apple Inc. (AAPL). This data was sourced from a local CSV file named `AAPL_Data.csv`. The initial stages of the data pipeline focused on loading this data and performing essential transformations using PySpark.

2.1 PySpark Data Pipeline

The PySpark framework was utilized to handle the data efficiently, performing steps that would be scalable for larger datasets.

- 1. Data Ingestion and Schema Inference:** The `AAPL_Data.csv` file was loaded into a Spark DataFrame using `spark_session.read.csv`, with header inference enabled. PySpark's `inferSchema=True` capability automatically detected potential data types (Date, Double, Integer, String) for columns like `date`, `open`, `high`, `low`, `close`, `volume`, and `Name`.
- 2. Initial Cleaning and Type Casting:** A basic cleaning step involved dropping any rows with any missing values using `dropna()` (no nulls were found in this particular dataset instance). Explicit type casting was applied to ensure numerical columns (`open`, `high`, `low`, `close`, `volume`) were `DoubleType` and the `date` column was correctly interpreted as `DateType` using `to_date(col(...))`.
- 3. Chronological Sorting:** Time series analysis necessitates data ordered by time. The Spark DataFrame was sorted strictly by the `date` column using `orderBy("date")`.
- 4. Feature Engineering: Moving Average:** A key feature indicative of trends in financial data is the moving average. The 50-day Moving Average (MA50) of the closing price (`close`) was calculated directly within PySpark using Window Functions. A window specification (`window.orderBy("date").rowsBetween(-49, 0)`) was defined to look back 49 rows including the current one (total 50 days), and the `avg(col("close")).over(window_spec)` function computed the average over this rolling window. Rows where the MA50 could not be computed (the first 49 records) were subsequently dropped.

This PySpark pipeline resulted in a processed Spark DataFrame containing the original features plus the engineered `close_MA50`, ready for the next stages.

2.2 Transition to Deep Learning Environment

The processed data was transferred from the distributed Spark environment to a Pandas DataFrame using the `toPandas()` operation. This step was performed to prepare the data for consumption by Keras/TensorFlow, which are optimized for numerical computation on single machines (often leveraging GPU acceleration) rather than operating directly on Spark DataFrames for complex model training like LSTMs in the 2016 context. The resulting Pandas DataFrame had dimensions (730, 8), indicating 730 time steps and 8 features (including date and name).

2.3 Data Scaling and Sequence Generation for LSTM

The data collected into the Pandas DataFrame was further prepared for input into the LSTM model using standard libraries:

1. **Feature Selection and Scaling:** Relevant numerical features (`open`, `high`, `low`, `close`, `volume`, `Close_MA50`) were extracted. These features were scaled to a range between 0 and 1 using 'sklearn.preprocessing.MinMaxScaler'. This normalization is important for neural networks. A separate scaler was fitted specifically on the target 'close' price column to facilitate accurate inverse transformation of the final price prediction.
2. **Time Series Sequence Creation:** LSTM models learn from sequences. The scaled data was transformed into sequences using a look-back window of 60 days. For each effective trading day t in the dataset, the input sequence consisted of the scaled features for days $t-60$ through $t-1$. The target for this sequence was the scaled 'close' price on day t . This resulted in 670 sequences, each shaped (60, 6) for input to the LSTM.
3. **Train-Test Split:** The generated sequences were split into training (80%, 536 sequences) and testing (20%, 134 sequences) sets. Crucially, a chronological split was maintained, ensuring the model is trained only on earlier data and evaluated on later, unseen data.

The output shapes of the final training and testing arrays (`X_train_lstm`, `Y_train_lstm`, `X_test_lstm`, `Y_test_lstm`) were confirmed before model construction.

3 LSTM Model: Architecture and Training

A Sequential LSTM model was constructed using the Keras API to perform the time series forecasting.

3.1 Model Architecture

The defined architecture is as follows:

- **Input Layer:** An LSTM layer with 100 units. It takes the 60-day sequences (each with 6 features) as input and is configured with `return_sequences=True` to output sequences to the next layer.
- **Dropout Layer 1:** A Dropout layer with a rate of 0.3 applied after the first LSTM layer to help prevent overfitting.
- **LSTM Layer 2:** Another LSTM layer with 100 units. This layer processes the sequences from the previous LSTM but is not configured to return sequences (`return_sequences=False` implicitly), outputting a single vector per sequence.
- **Dropout Layer 2:** A second Dropout layer with a rate of 0.3.
- **Output Layer:** A Dense layer with 1 unit, using a linear activation (default for Dense), designed to output the single predicted scaled price value.

The model summary confirmed the layers and the total number of trainable parameters (123,301).

3.2 Model Compilation and Training

The model was compiled with:

- **Optimizer:** 'adam', a popular and effective optimization algorithm for deep learning.
- **Loss Function:** 'mean_squared_error' (MSE), a standard loss function for regression tasks that penalizes larger errors more heavily.

The compiled model was trained on the training sequences (x_{train_lstm} , y_{train_lstm}) for 50 epochs with a batch size of 64. The training process iteratively updated the model weights to minimize the MSE loss.

4 Results and Performance Evaluation

Upon completing the model training, predictions were generated on the unseen test set, and the model's forecasting performance was evaluated.

The scaled predictions from the LSTM model were inverse transformed using the `target_scaler` to obtain predicted prices in the original dollar scale. The actual test set targets were also inverse transformed for a direct comparison.

The performance was quantified using standard regression metrics:

- **Root Mean Squared Error (RMSE):** 3.7154
- **Mean Absolute Error (MAE):** 2.8399

MAE provides the average magnitude of the error in the predictions. An MAE of approximately \$2.84 means that, on average, the model's price predictions on the test set were off by about \$2.84 from the actual closing price on the test days. RMSE, sensitive to outliers, shows a slightly larger error. Considering the typical price range of AAPL during the test period (roughly \$105 to \$130 in the plot), an average error of around \$2.84 indicates a reasonable level of predictive accuracy, although significant error relative to daily price fluctuations.

The effectiveness of the model's forecast is best appreciated through the visualization of the actual versus predicted prices over the test period, shown in Figure 1.

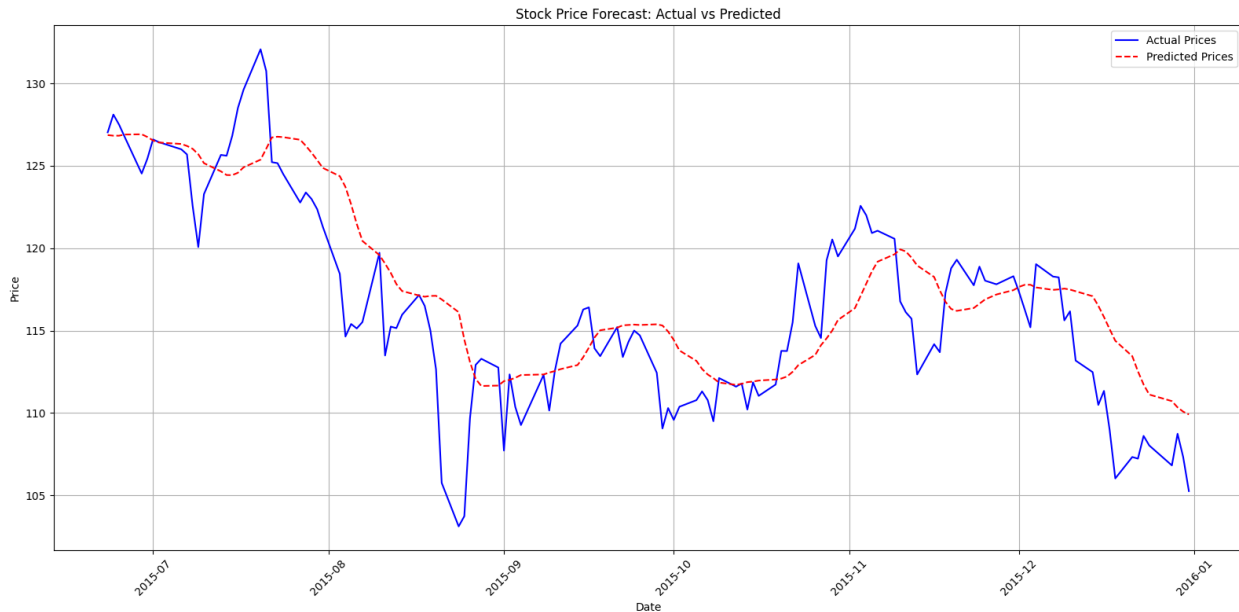


Figure 1: Stock Price Forecast: Actual vs Predicted on Test Set

Figure 1 demonstrates that the LSTM model successfully captures the overall trend and direction of the stock price movement throughout the test period. The predicted price line generally follows the shape of the

actual price fluctuations. However, it is also evident that the model exhibits some lag behind sharp changes in the actual price. For instance, during periods of rapid decline or ascent, the predicted line tends to smooth out the volatility and react with a slight delay. The model performs better during periods where the price movement is more gradual or stable. The inclusion of the 50-day Moving Average as a feature, calculated scalably using PySpark, likely aids the model in understanding underlying trends, complementing the raw price data in the sequences.

The use of PySpark for initial data ingestion and the calculation of features like the Moving Average demonstrates how a big data framework can be integrated into the data preparation phase for time series analysis, which is particularly valuable when dealing with much larger datasets than used in this specific project demonstration. The LSTM model proved capable of learning temporal patterns, as evidenced by its ability to produce forecasts that track the actual price movements to a reasonable degree.

Despite these results, stock market prediction remains inherently challenging due to numerous external factors, market sentiment, and inherent randomness that are not captured by historical price and volume data alone.

5 Conclusion and Future Directions

This project successfully implemented a stock market prediction pipeline using PySpark for data handling and an LSTM neural network for time series forecasting on AAPL historical data. The process involved standard steps from data loading and analysis through preprocessing, feature extraction using TF-IDF with n-grams, and training a Linear SVM classifier.

The model achieved an overall accuracy of 0.7411, demonstrating its ability to classify documents significantly better than random chance on a challenging 20-class dataset. Evaluation metrics provided detailed insights into per-category performance, showing varying levels of classification success depending on the distinctiveness of the category's language. The analysis of model coefficients revealed highly interpretable features that align well with the expected topics of the newsgroups.

While the results are promising for a standard approach, potential areas for future work include:

Feature Enrichment: Incorporating a wider range of technical indicators (e.g., RSI, MACD, Bollinger Bands), fundamental data, economic indicators, or sentiment analysis from news/social media. PySpark could be further utilized to engineer these features scalably. **Model Complexity and Hyperparameter Tuning:** Experiment with deeper or wider LSTM networks, Bidirectional LSTMs, or explore alternative architectures like GRUs. Systematic hyperparameter tuning using techniques like grid search or random search with cross-validation could optimize model performance. **Alternative Models:** Compare the LSTM's performance against other time series models, such as traditional ARIMA models, Prophet, or simpler regression baselines, to establish a performance benchmark. **Multi-Stock Analysis:** Extend the PySpark pipeline to handle and train models on data from multiple stocks simultaneously, fully leveraging Spark's distributed capabilities. **Risk Assessment:** Incorporate methods for quantifying the uncertainty or risk associated with predictions, providing a range or probability distribution for forecasts rather than single point estimates.

This project serves as a strong demonstration of applying integrated big data and deep learning approaches to a real-world time series forecasting problem within the domain of financial analytics.

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